Assignment (4.2)

Consider the following table: Table: ACCOUNT_TXN

Acc_ID	Holder_Name	Balance
101	Aman	5000
102	Bhavna	7000
103	Chirag	6000

Perform the following TCL-based SQL script and write the output after each SELECT statement:

```
INSERT INTO ACCOUNT_TXN VALUES (104, 'Diya', 4000);

SAVEPOINT S1;

UPDATE ACCOUNT_TXN SET Balance = Balance + 1500 WHERE Acc_ID = 101;

UPDATE ACCOUNT_TXN SET Balance = Balance - 1000 WHERE Acc_ID = 102;

SAVEPOINT S2;

DELETE FROM ACCOUNT_TXN WHERE Acc_ID = 103;

INSERT INTO ACCOUNT_TXN VALUES (105, 'Eshan', 3000);

SAVEPOINT S3;

UPDATE ACCOUNT_TXN SET Holder_Name = 'Bhavna Sharma' WHERE Acc_ID = 102;

SELECT * FROM ACCOUNT_TXN;
```

```
ROLLBACK TO S2;

SELECT * FROM ACCOUNT_TXN;

INSERT INTO ACCOUNT_TXN VALUES (106, 'Farah', 4500);

SAVEPOINT S4;

UPDATE ACCOUNT_TXN SET Balance = Balance + 500 WHERE Acc_ID IN (101,104);

SELECT * FROM ACCOUNT_TXN;

ROLLBACK TO S4;

SELECT * FROM ACCOUNT_TXN;

COMMIT;
```

Student Tasks

Write the output of all 4 SELECT statements in tabular format.

Identify which rows are permanently saved after COMMIT.

Mention which operations were undone by:

ROLLBACK TO S2

ROLLBACK TO S4

Write the final state of the table after COMMIT.

Additional Challenge (Conceptual)

Answer the following:

What is the difference between ROLLBACK and ROLLBACK TO SAVEPOINT?

Can we rollback to S3 after ROLLBACK TO S2? Why/Why not?

What happens if COMMIT is executed before creating any savepoint?

```
DROP TABLE IF EXISTS ACCOUNT_TXN; -- ASSIGNMENT 4.2

CREATE TABLE ACCOUNT_TXN (
   Acc_ID INT PRIMARY KEY,
   Holder_Name VARCHAR(50),
   Balance INT
) ENGINE=InnoDB;

INSERT INTO ACCOUNT_TXN VALUES
(101,'Aman',5000),
(102,'Bhavna',7000),
(103,'Chirag',6000);

SET autocommit = 0;

INSERT INTO ACCOUNT_TXN VALUES (104, 'Diya', 4000);

SAVEPOINT S1;

UPDATE ACCOUNT_TXN
```

```
SET Balance = Balance + 1500
WHERE Acc_ID = 101;

UPDATE ACCOUNT_TXN
SET Balance = Balance - 1000
WHERE Acc_ID = 102;

SAVEPOINT S2;

DELETE FROM ACCOUNT_TXN
WHERE Acc_ID = 103;

INSERT INTO ACCOUNT_TXN VALUES (105, 'Eshan', 3000);

SAVEPOINT S3;

UPDATE ACCOUNT_TXN
SET Holder_Name = 'Vanshika Dixit'
WHERE Acc_ID = 102;

SELECT * FROM ACCOUNT_TXN;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Acc_ID	Holder_Name	Balance
101	Aman	6500
102	Vanshika Dixit	6000
104	Diya	4000
105	Eshan	3000
NULL	NULL	NULL

```
ROLLBACK TO S2;

SELECT * FROM ACCOUNT_TXN;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Acc_ID	Holder_Name	Balance
101	Aman	6500
102	Bhavna	6000
103	Chirag	6000
104	Diya	4000
NULL	NULL	NULL

INSERT INTO ACCOUNT_TXN VALUES (106, 'Farah', 4500);

SAVEPOINT S4;

UPDATE ACCOUNT_TXN

SET Balance = Balance + 500

WHERE Acc_ID IN (101,104);

SELECT * FROM ACCOUNT_TXN;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Acc_ID	Holder_Name	Balance
101	Aman	7000
102	Bhavna	6000
103	Chirag	6000
104	Diya	4500
106	Farah	4500
NULL	NULL	NULL

ROLLBACK TO S4;

SELECT * FROM ACCOUNT_TXN;

COMMIT;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Acc_ID	Holder_Name	Balance
101	Aman	6500
102	Bhavna	6000
103	Chirag	6000
104	Diya	4000
106	Farah	4500
NULL	NULL	NULL